

Exp. No. 5

Design and Simulate 3-Bit Ring Counter using Verilog code in Xilinx

Aim

To design and simulate a 3-bit Ring Counter using Verilog HDL in the Xilinx ISE/Vivado environment.

Verilog Code

```
module ring_counter (
    input clk,
    input reset,
    output reg [2:0] q
);

always @(posedge clk or posedge reset)
begin
    if(reset)
        q <= 3'b001;           // Initial state
    else
        q <= {q[1:0], q[2]};    // Circular left shift
    end
endmodule
```

Testbench Code

```
`timescale 1ns / 1ps
module ring_counter_tb;

    reg clk;
    reg reset;
    wire [2:0] q;

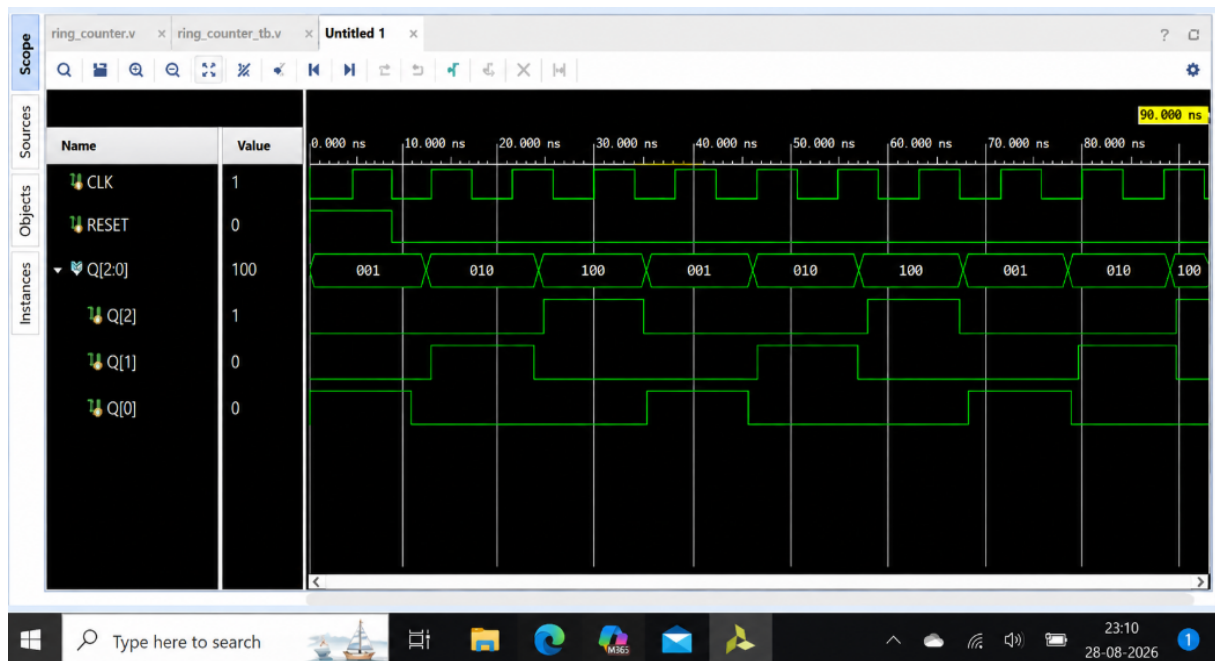
    ring_counter uut (
        .clk(clk),
        .reset(reset),
        .q(q)
    );

    // Clock Generation
    always #5 clk = ~clk;

    initial
    begin
        clk = 0;
        reset = 1;
        #10;
        reset = 0;
        #80;
        $finish;
    end

endmodule
```

Output (Simulation Waveform)



Result: The 3-bit Ring Counter was designed and simulated successfully. On reset, Q is initialized to 001, and on every rising edge of the clock, the single '1' bit circulates through the register (001 → 010 → 100 → 001 ...), as verified in the simulation waveform.